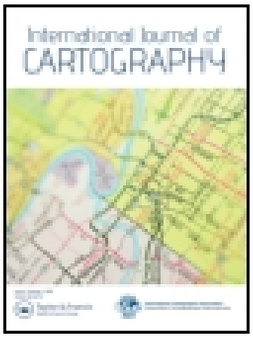

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# Simplification of polylines by segment collapse: minimizing areal displacement while preserving area

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**ABSTRACT**

This paper reports on a new Area Preserving Segment Collapse (APSC) algorithm for simplifying polygonal boundaries while preserving the polygonal area at simplified target scales and minimizing areal displacement. A general segment collapse algorithm is defined by iteratively collapsing segments to Steiner points in priority order, guided by placement and displacement functions. The algorithm is specified by defining functions that minimize areal displacement under the constraint that the areas of adjoining polygons are preserved exactly. Self-intersections can be avoided by testing for intersections with two new line segments associated with each segment collapse operation. The paper demonstrates simplification results for a sample of 10 lakes formed from alpine, Karst, glacial and arid desert processes as well as artificial dams. APSC results are compared with three other simplification routines and evaluated for area preservation, linear and areal displacement, complexity and introduction of boundary self-intersections. Results confirm that the APSC algorithm preserves area exactly and indicate that it outperforms the other tested algorithms for minimizing areal displacement while producing reasonably low measures of linear displacement. Self-intersections can occur more commonly with APSC than other algorithms but are avoided with the proposed topology check. The APSC algorithm additionally preserves polygon complexity better than other tested algorithms.

**RÉSUMÉ**

Ce papier présente un nouvel algorithme appelé APSC (Area Preserving Segment Collapse) pour simplifier les limites polygonales tout en préservant les surfaces à une échelle cible et en minimisant la surface de déplacement. Nous définissons un algorithme de réduction de segment en regroupant de façon récursive des segments aux points de Steiner par ordre de priorité et guidé par des fonctions de positionnement et de déplacement. L'algorithme est spécifié en définissant des fonctions qui minimisent la surface de déplacement sous la contrainte que les surfaces des polygones adjacents soient exactement préservées. Les intersections de la ligne avec elle-même peuvent être évitées en testant les intersections avec deux nouveaux segments de

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ligne associés à chaque opération de réduction de segment. Le papier présente des résultats de simplification pour un échantillon de 10 lacs formés à partir de processus alpin, karstique, glacier et de désert aride ainsi que des barrages artificiels. Les résultats de l'APSC sont comparés avec trois autres algorithmes de simplification et sont évalués sur des critères de préservation de surface, de déplacement linéaire et surfacique, de complexité et de création d'intersection de la ligne avec elle-même. Les résultats confirment que l'algorithme APSC préserve exactement la surface et montre qu'il surpasse les autres algorithmes testés pour minimiser le déplacement surfacique tout en produisant des valeurs raisonnablement faibles pour le déplacement linéaire. L'auto-intersection peut se produire plus souvent avec l'APSC qu'avec d'autres algorithmes mais ces intersections sont évitées grâce à un contrôle topologique. De plus l'algorithme APSC préserve mieux la complexité des polygones que les autres algorithmes testés.

## Introduction

Developing simplified representations of a two-dimensional polyline is an important problem in cartographic data analytics where datasets must be integrated a cross spatial resolutions. This problem is generally referred to as line simplification and is driven by preservation of global shape as well as specific analytic properties such as positional accuracy and high-frequency detail. However, the distinction between linear features and polygon boundaries is rarely considered. Polygonal features differ fundamentally from linear features in that they represent areal regions covering a portion of the Earth's surface. As such, they lend themselves naturally to the additional objective of preserving area across scales of representation.

Only a limited number of studies have been conducted on area-preserving polyline simplification techniques, and these have focused on theoretical aspects (Bose et al., 2006) and the development of algorithms for schematic simplification (Meulemans, van Renssen, & Speckmann, 2010) and smoothing (Tutić & Lapaine, 2009). Although areal displacement has been used both to guide (Visvalingam & Whyatt, 1992) and to measure the quality (McMaster, 1986) of line simplification algorithms, the directional symmetry or asymmetry of displacement is rarely considered. Shen, Ai, and He (2018) recently compared several simplification algorithms on lake boundaries for effects on area and positive and negative displacement area, but no simplification constraints were applied to control these effects. We argue that displacement symmetry leads directly to area preservation, bringing both intrinsic value as an objective function in and of itself as well as utility in constraining solutions to minimize overall deviation from the original polygonal boundary.

The fundamental value of area preservation derives from perceptual and analytic requirements in various domains for accurate database and map representations of area. On thematic maps, it is important to be able to perceive the relative density of a variable, and calculation of attributes such as population density depends on accurate retention of area in simplified polygon datasets. Where line simplification is used on a cartogram, area preservation is required to retain the accuracy of the cartogram. The

analytical importance of area extends to nearly all forms of geospatial analysis, for example to domains such as hydrologic modeling where the surface area of lakes and streams determines evaporation rates and geochemical exchanges between the air and water below it. Where such modeling extends across large study areas that warrant data reduction and boundary simplification, polygonal areas must be preserved to support accurate computations.

Beyond the intrinsic value of preserving area for specific perceptual and analytic purposes, there are logical arguments as to why area preservation might generally lead to better preservation of global shape in reduced-scale representations of any polyline. For instance, If the areal displacement is larger on one side of the line than on the other, a better simplified representation could be obtained by shifting the line towards the direction of greater displacement. Since this logic applies to any line, 'area' preservation might be a useful criterion even for lines that do not bound areal regions, such as single-line rivers and roads.

To explore these ideas further, a novel area-preserving polyline simplification algorithm is developed. The algorithm works by iteratively collapsing segments to points and is referred to as Area-Preserving Segment Collapse (APSC). The concept of segment collapse has been utilized in previous algorithms with explicitly visual aims of smoothing (Tutić & Lapaine, 2009) and collapsing visually non-discernible vertices (Raposo, 2013). In contrast, APSC is driven by analytical error-minimization objectives. Specifically, selection and collapse are controlled by the simultaneous goals of minimizing areal displacement and balancing area preservation across both sides of a simplified boundary. Procedurally, APSC is similar to the Visvalingam and Whyatt (1992) Effective Area (VEA) algorithm, but instead of removing a single vertex at each step, two vertices are removed and one new vertex is added.

To avoid confusion, we distinguish between two related but distinct concepts that are both used to guide the APSC algorithm. Area preservation is a term that refers to the precise equivalence of area computation prior to and following simplification processing. Areal displacement, in contrast, refers to local boundary shifts during processing. All line simplification algorithms result in some degree of areal displacement between the original and simplified polyline. Nevertheless, it is possible to avoid a global reduction or enlargement in the size of a polygon, and thus preserve area exactly, by balancing displacement on either side of a polygon boundary. The resulting area at target scales precisely matches area at the source scale, even though some areal displacement occurs locally.

The segment collapse algorithm also differs from many commonly used line simplification algorithms in that it introduces new vertices (Steiner points). Indeed, it is unlikely that any line simplification algorithm can preserve area exactly without using Steiner points (Bose et al., 2006). The introduction of Steiner points does not appear to obstruct preservation of critical points, which tend to be situated where linear features change direction and are acknowledged to be important for shape recognition (Attneave, 1954). Thus, Steiner points can preserve a geometric property (area) needed for analytical purposes, without corrupting the visual property (angular change) needed for cartographic display.

The paper proceeds as follows. The next section reviews line simplification algorithms, highlighting those that preserve area and/or introduce Steiner points. Section three describes and illustrates the proposed segment collapse algorithm and strategy to minimize areal displacement while meeting the area-preserving objective, as well as a strategy

to prevent self-intersection. Section four presents the experimental design, detailing the comparison of APSC with VEA, Ramer-Douglas-Peucker (RDP) and Raposo Spatial Means (RSM) algorithms. The algorithms are tested on 10 lakes from across the conterminous United States with varying spatial structure and geomorphic origin. The results of these experiments are presented in section five. The last section discusses advantages and disadvantages of the proposed algorithm and potential directions for future research.

## Background

### *Line simplification*

Numerous line simplification algorithms have been proposed and implemented. Published reviews categorize algorithms based on the types of errors that are introduced (McMaster, 1986), various conceptual models that guide processing (McMaster & Shea, 1992), a comparison of assumptions underlying generalization processing (Buttenfield, 1985), chronologies of algorithm development (Li, 2006, 2007) and inventories of generalization operators (Stanislawski, Buttenfield, Bereuter, Savino, & Brewer, 2014). Interested readers are encouraged to explore these resources for comprehensive discussions about specific approaches and algorithms.

Several algorithms are commonly encountered in GIS software platforms. The RDP algorithm simplifies a feature by retaining a subset of vertices that lie further than the tolerance distance from a hierarchy of baselines composed from the feature's vertices (Douglas & Peucker, 1973; Ramer, 1972). The simplification proceeds hierarchically, effectively prioritizing coordinate positions for progressively more aggressive simplifications. This algorithm is useful for simplifying human-made linear features whose angularity is deemed important, such as building polygons, roads or administrative boundaries. It carries the added advantage of preserving a subset of original coordinates at every level of simplification. The Bend-Simplify algorithm (Wang & Müller, 1998) moves along a line and eliminates points that demarcate small curves and bends while preserving larger trending shapes. The algorithm attempts to simulate manual line simplification by applying rule-based operations on bend features that would be identified and processed (i.e. removed or merged) by a human cartographer. The Bend-Simplify algorithm is typically used for naturally occurring features, such as stream channels and lakes (Gary, Wilson, Archuleta, Thompson, & Vrabel, 2010). The VEA method attempts to minimize areal displacement by forming triangles from triplets of sequential points and replacing the smallest triangles with a straight-line segment by eliminating the middle point (Visvalingam & Whyatt, 1992). While areal displacement is effectively reduced, the algorithm does not preserve area precisely.

### *Steiner points*

The above algorithms all produce simplified polylines that consist of a subset of the original polyline vertices, with no 'Steiner' points. The concept of Steiner points comes from the discipline of computational geometry and refers to a point or set of points that are introduced when solving a geometric optimization problem to improve upon solutions based only on the original point set (de Berg, van Kreveld, Overmars, & Schwarzkopf,

2000). Steiner points have been used, for example, to develop triangulations with better properties such as larger minimum angle and shorter total line length (Eppstein, 1994). In the context of line simplification, the flexibility afforded by Steiner points may enable solutions with less displacement than popular algorithms that do not employ Steiner points. For example, Cromley (1992) presents an algorithm generating a principal axis or regression line through subsets of vertices, using Steiner points to reduce a line to a given tolerance using fewer vertices than required by the RDP algorithm.

Several algorithms utilizing Steiner points operate by enclosing a small number of original coordinates within a regular polygon and then reducing them to a single representative (Steiner) point, such as the centroid of the regular polygon or the spatial mean of the vertices located within that polygon. Li and Openshaw (1992) overlay a regular square grid onto a polyline and collapse vertices within each grid cell to a single point. Raposo (2013) improves on the methods of Li and Openshaw by using a hexagonal tesselation instead of a square grid. In the RSM algorithm, coordinates falling within each hexagon are replaced with the spatial mean of those coordinates, which is likely a Steiner point when more than one point falls within a hexagon. RSM has the added advantage of obtaining significantly lower horizontal displacement errors than occur with Li and Openshaw's (1992) method.

It is worthwhile to examine the general case for Steiner points, and how the inclusion of points that are not in the original coordinate set can lead to a better representation. Consider the case of a polygon with a large number of vertices that approximates a circle, and the problem of representing this feature with a simpler polygon containing only six vertices (Figure 1). An optimal simplification will be a regular hexagon. If the $n$ vertices are constrained to the original polygon (Figure 1(a)), it will be an inscribed hexagon, and 17.3% of the original polygon will fall outside the simplified shape. On the other hand, if Steiner points are allowed, one can construct a hexagon with an equal area to the original polygon. In this case, 3.7% of the original polygon will fall outside the hexagon, and an equal area of the hexagon will fall outside the original polygon. Thus the total areal displacement will be 7.4%, an improvement of 57% from the optimal solution without Steiner points.

Note that the introduction of Steiner points not only reduces the overall displacement area but also balances it on either side of the original polygon boundary, thus preserving the area of the polygon.

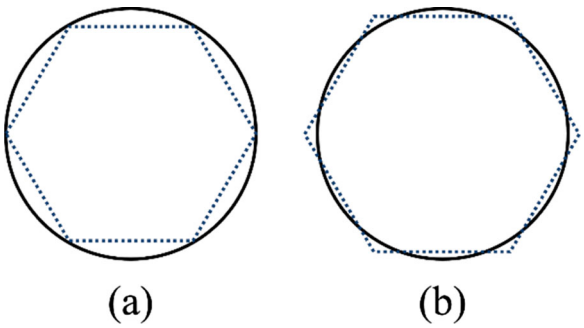


**Figure 1.** Optimal simplification of a polygon simplifying a circle to six vertices without (a) and with (b) Steiner points.

### Handling topological errors

Most line simplification algorithms can produce topological errors such as self-intersections, and various strategies have been proposed to avoid these errors. These can be divided into strategies to prevent topological errors from occurring during the line simplification process (e.g. Kulik, Duckham, & Egenhofer, 2005; Saalfeld, 1999) and strategies to detect and fix topological errors after simplification (e.g. Lee & Hardy, 2005; Raposo, 2013). Of most relevance to our work is the observation by Kulik et al. (2005) that the triangle-criterion, derived from mathematical topology, enables a simple topology check for any algorithm that works by sequential vertex elimination. The triangle-criterion states that when a vertex is eliminated from a polyline, topological errors will be introduced only if there exists a point lying in the triangle formed by that vertex and its neighbors. Thus, for any sequential elimination algorithm, a simple point-in-triangle test can be used to identify and avoid topological errors associated with eliminating each vertex.

### Area preservation

Only a few studies have investigated the preservation of area in polyline simplification. The lack of algorithms that preserve area is an inevitable consequence of the fact that most line simplification algorithms retain subsets of the original vertices, since the general problem of determining if a line can be simplified to preserve adjoining areas is intractable for large polylines (i.e. NP-hard) when only existing vertices are used (Bose et al., 2006). Indeed, on a closed convex polygon such as shown in Figure 1, preserving adjoining areas without using Steiner points is demonstrably impossible. Thus, the few algorithms that have been developed to preserve area either are inexact or introduce Steiner points.

Bose et al. (2006) develop an algorithm to preserve area to within a specified tolerance without using Steiner points. Their algorithm can only be applied, however, to so-called monotone chains, which are polylines or sections of polylines that can be rotated such that they cross any given *x*-coordinate only once. Meulemans et al. (2010) present an exact area-preserving schematization of polygon tessellations that uses Steiner points. Their schematization is rectilinear and provides effective simplification of artificial features such as buildings, but the algorithm is not suitable for lakes and other natural features.

Tutić and Lapaine (2009) present an area-preserving line smoothing algorithm that is most applicable to the cartographic generalization of natural features, and that shares some characteristics of the APSC algorithm. Collapsing individual segments to Steiner points, they demonstrate that area is preserved if these Steiner points are located on a specific line whose equation can be computed from the coordinates of the vertices of the collapsed segment and immediately adjacent segments. While our proposed algorithm utilizes this result, it differs from the algorithm of Tutić and Lapaine (2009) in three ways. First, their algorithm traverses the input polyline sequentially, whereas the one proposed here searches for the optimal segment to collapse at each step. Secondly, Tutić and Lapaine collapse only zig-zag sections (S-curves) in which two consecutive polyline segments turn in different directions, and they do not handle curves where two consecutive segments curve in the same direction (C-curves). This appears to substantially restrict the number of vertices that can be eliminated with their algorithm. APSC does simplify both S-

curves and C-curves. Finally, in choosing the placement of new Steiner points, Tutić and Lapaine seek to minimize sharp angles, resulting in an algorithm that effectively smooths the original polyline. In contrast, the proposed APSC algorithm minimizes areal displacement, and thereby produces a simplified polyline that is similar in character to the original polyline.

## Methods

### *General segment collapse algorithm*

A general segment collapse algorithm can be defined by iteratively collapsing segments to points, selecting at each iteration the collapse operation that will result in the least displacement. A collapse operation begins with a four-vertex sequence $\overline{ABCD}$ and involves removing the interior segment $\overline{BC}$ and replacing it with a single vertex $E$ (Figure 2).

Two functions are required to fully specify the algorithm. First, a *placement* function determines, for a given sequence of vertices $\overline{ABCD}$, the location to place the new vertex $E$. Second, a *displacement* function measures the associated degree of displacement between $\overline{ABCD}$ and $\overline{AED}$. The *placement* function is used to create a list of potential collapse operations for the current set of line segments, and to update this list after each operation is performed. The *displacement* function is used to prioritize the order in which segments are collapsed. The process continues until a stopping condition is met. Pseudo-code for simplifying a line $L$ to a target areal displacement tolerance $t$ is as follows:

**function segment_collapse(L, t)**

```
worklist ← empty list
for all consecutive sequences ABCD in L:
    E ← placement (ABCD)
    δ ← displacement(ABCD → AED)
    add [ABCD, E, δ] to worklist
while worklist is not empty:
    [ABCD, E, δ] ← item with min δ in worklist
    if δ < t:
        collapse BC to E
        with worklist:
```

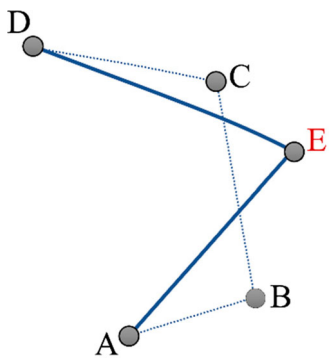


**Figure 2.** Segment collapse. Segment $\overline{BC}$ on the original (dashed) feature is replaced by vertex $E$ on the simplified (solid) feature.

```
        remove all 4-vertex sequences involving B or C
        add all 4-vertex sequences involving E
    else exit
```

The algorithm can also be modified to stop once a target number of vertices is reached, rather than a target areal displacement tolerance. Alternatively, cumulative areal displacement can be tracked and used to parameterize the algorithm. The algorithm is general in the sense that different placement and displacement functions can be defined to meet different objectives.

As a specific instance of the general algorithm, we define an *area-preserving segment collapse* (APSC) algorithm by specifying the displacement function as *areal displacement*, i.e. the total area enclosed by $\overline{ABCD}$ and $\overline{AED}$ (McMaster, 1986), and by defining a placement function that both (a) preserves area on each side of the original line, and (b) minimizes areal displacement under the area-preserving constraint.

## *Area preservation*

The conditions for preserving area while collapsing a segment to a Steiner point are given in Tutić and Lapaine (2009). Consider the four-vertex sequence $\overline{ABCD}$ in Figure 3(a). We seek a new point $E$ such that the areas on each side of $\overline{AED}$ are equal to the areas on each side of $\overline{ABCD}$. It is sufficient to measure the areas between these polyline sections, separating them into areas to the left (L) and right (R) of the original polygon as shown in Figure 3. The area preservation condition dictates that these two areas must be equal.

Let $Area^{\circlearrowleft}(\Delta ijk)$ denote the area of the triangle with vertices *i, j* and *k* using the standard clockwise-sequence shoelace formula or Surveyor's formula (Braden, 1986), i.e. measured such that the area is negative if the vertices are sequenced counterclockwise. The area preservation condition can be expressed as follows:

$$Area^{\circlearrowleft}(\Delta ACB) + Area^{\circlearrowleft}(\Delta ADC) = A^{\circlearrowleft}(\Delta ADE) \tag{1a}$$

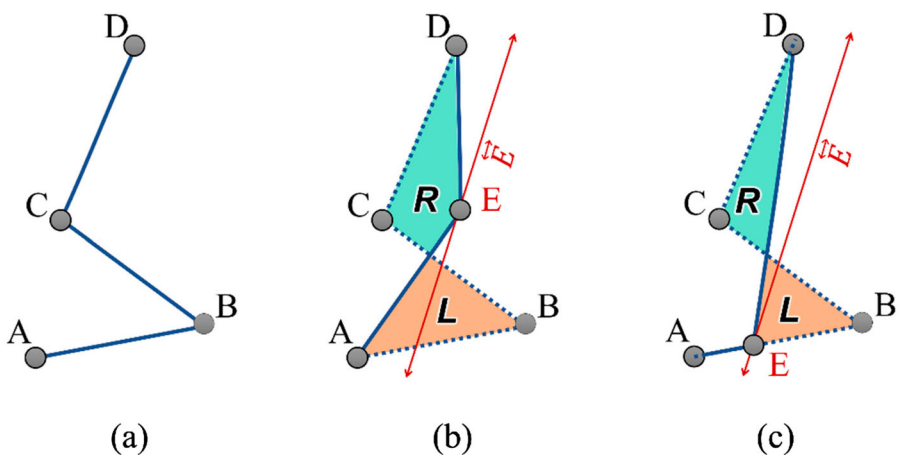


**Figure 3.** Area preservation condition. (a) a four-vertex sequence on the original polygon, (b) placement of new vertex E to maximize angle ∡AED while preserving area, (c) placement of new vertex E to minimize areal displacement while preserving area.

This simplifies to the following equation:

$$ax_E + by_E + c = 0, \tag{1b}$$

where:

$$a = y_D - y_A,$$

$$b = x_A - x_D,$$

$$c = -y_B x_A + (y_A - y_C)x_B + (y_B - y_D)x_C + y_C x_D.$$

Equation (1) defines a line that is parallel to $\overline{AD}$, which we denote as $\overleftrightarrow{E}$ (Figure 3(b,c)). This line represents the set of possible locations of the vertex $E$ that would meet the area preservation condition. Intuitively, the distance from $\overline{AD}$ to $\overleftrightarrow{E}$ represents the height of triangle $\Delta$ADE whose base is $\overline{AD}$. Figure 3(b,c) illustrates two possible solutions that preserve area, since $E$ is located on $\overleftrightarrow{E}$ in both figures.

## *Minimizing areal displacement*

Since the area is preserved, placement of $E$ anywhere on $\overleftrightarrow{E}$ will result in equal areal displacement to the left and right of the original polyline. However, the magnitude of this displacement will vary depending on where on $\overleftrightarrow{E}$ the new vertex is placed. It is natural to seek the location of $E$ that will minimize total areal displacement, thus meeting the second objective mentioned above. This objective was mentioned but not solved by Tutić and Lapaine (2009), who instead chose to maximize the angle $\measuredangle$AED. Figure 3 illustrates the difference between the angle-maximizing solution (Figure 3(b)) and the solution that minimizes areal displacement (Figure 3(c)).

To develop the areal-displacement-minimizing solution, we consider regular topological configurations of the vertices $B$ and $C$ and the lines $\overline{AD}$ and $\overleftrightarrow{E}$. Specifically, we show that the minimum areal displacement is obtained by placing $E$ at the intersection of $\overleftrightarrow{E}$ with either $\overline{AB}$ or $\overline{CD}$ depending on the topological configuration of $B$, $C$, $\overleftrightarrow{E}$ and $\overline{AD}$. Excluding singularities, three configurations are possible (Figure 4):

1. vertices $B$ and $C$ are on opposite sides of both $\overline{AD}$ and $\overleftrightarrow{E}$ (Figure 4(a)),
2. vertices $B$ and $C$ are on the same side of $\overline{AD}$ but different sides of $\overleftrightarrow{E}$ (Figure 4(b)), or
3. vertices $B$ and $C$ are located between $\overline{AD}$ and $\overleftrightarrow{E}$ (Figure 4(c)).

Vertices $B$ and $C$ cannot both lie on the same side of both $\overline{AD}$ and $\overleftrightarrow{E}$ because if they did, then the area preservation condition would not be met.

The location of $E$ that minimizes areal displacement for each configuration is shown in Figure 4. Consider the first configuration, with vertices $B$ and $C$ on opposite sides of both $\overline{AD}$ and $\overleftrightarrow{E}$ (Figure 4(a)). If $B$ is on the same side of $\overline{AD}$ as $\overleftrightarrow{E}$ (as shown) then the optimal location for $E$ is the intersection of $\overline{AB}$ and $\overleftrightarrow{E}$, otherwise, it is the intersection of $\overline{CD}$ and $\overleftrightarrow{E}$. To confirm that this is optimal, consider any alternative location $E'$ on $\overleftrightarrow{E}$ and the corresponding displacement areas $L'$ and $R'$ to the left and right of the original (dashed) line (Figure 5). For any $E'$ below $E$, $L'$ includes $L$ as a proper part and is, therefore, larger than $L$ (Figure 5(b)). If $E'$ is above $E$ two cases must be considered. For $E'$ such that the intersection of $\overline{AE'}$ and $\overline{BC}$ is closer to $B$ than the intersection of $\overline{ED}$ and $\overline{BC}$ (Figure 5(c)), $R'$ will

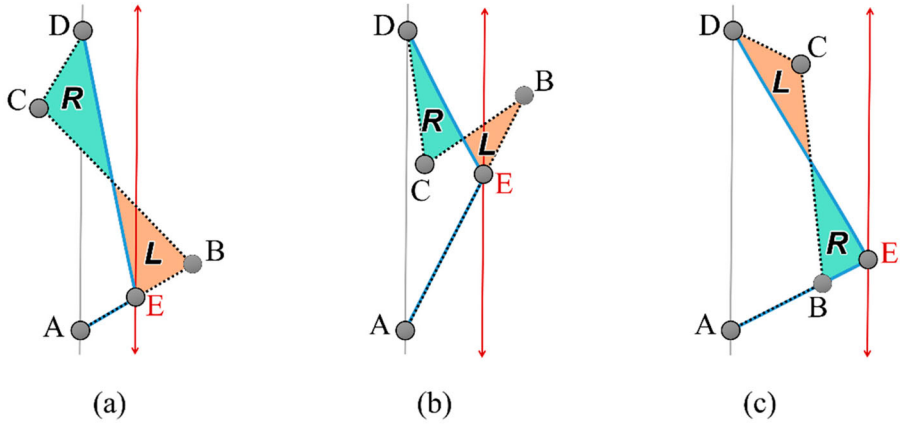


**Figure 4.** Possible configurations of $B$, $C$, $\overline{AD}$ and $\overleftrightarrow{E}$. (a) $B$ and $C$ are on opposite sides of both $\overline{AD}$ and $\overleftrightarrow{E}$; (b) $B$ and $C$ are on the same side of $\overline{AD}$ but opposite sides of $\overleftrightarrow{E}$; (c) $B$ and $C$ both lie between $\overline{AD}$ and $\overleftrightarrow{E}$. Optimal placement of $E$ is at intersection of $\overleftrightarrow{E}$ and $\overline{AB}$ if: (a) $\overleftrightarrow{E}$ and $B$ are on the same side of $\overline{AD}$; (b or c) $B$ is further than $C$ from $\overline{AD}$. Otherwise, optimal placement of $E$ is at intersection of $\overleftrightarrow{E}$ and $\overline{CD}$.

include $R$ as a proper part. On the other hand, for $E'$ such that the intersection of $\overline{AE'}$ and $\overline{BC}$ is closer to $C$ than the intersection of $\overline{ED}$ and $\overline{BC}$ (Figure 5(d)), $L'$ will include $L$ as a proper part. Since $L=R$ (by Equation (1)), in all cases the total areal displacement $L'+R'$ will be larger than $L+R$.

The logic shown in Figure 5(b,c) applies to the configuration in Figure 4(b) for any $E'$ above or below $E$. For the configuration in Figure 4(c), the logic shown in Figure 5(c) applies for any $E'$ below $E$. For $E'$ above $E$, $R' > R$ can be proven geometrically by splitting $R$ and $R'$ into two triangles along a vertical line through $B$, and noting that both their shared baseline and the sum of their altitudes will increase. Building on these observations, a common rule can be applied to the configurations in both 4b and 4c: if $B$ is further from $\overline{AD}$ than $C$ then the location of $E$ that minimizes areal displacement is the intersection of $\overline{CD}$ and $\overleftrightarrow{E}$, otherwise it is the intersection of $\overline{AB}$ and $\overleftrightarrow{E}$.

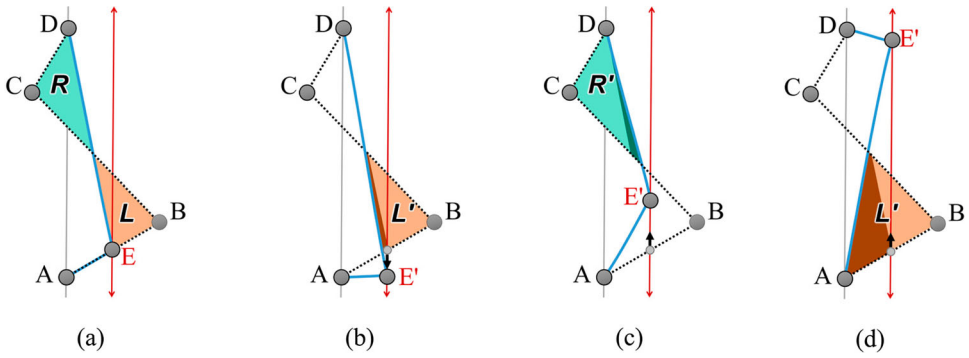


**Figure 5.** Illustration of the optimality of the placement of $E$ in Figure 4(a). (a) optimal placement at the intersection of $\overline{AB}$ and $\overleftrightarrow{E}$, (b) placement below the intersection of $\overline{AB}$ and $\overleftrightarrow{E}$, (c) placement above the intersection of $\overline{AB}$ and $\overleftrightarrow{E}$. Dark shaded areas in (b) and (c) signify the additional region of displacement added to $L$ and $R$, respectively, by moving $E$ from its position in (a).

Combining the above observations, a *placement* function for choosing the location of the new vertex $E$ to replace vertices $B$ and $C$ in a sequence of vertices $A$, $B$, $C$, $D$ can be expressed in pseudo-code as follows:

```
function placement(A,B,C,D)

Compute Ê using eq. 1.
if side(B, AD) = side(C, AD):
    if d(B, AD) > d(C, AD):
        return intersection(Ê , AB)
    Else if d(B, AD) < d(C, AD):
        return intersection(Ê , CD)
else:
    if side(B, AD) = side(Ê , AD):
        return intersection(Ê , AB)
    else:
        return intersection(Ê , CD)
```

Here, $d(x, L)$ denotes the perpendicular distance between point $x$ and line L, and side $side(x, \mathrm{L})$ denotes the topological position of a point $x$ (or a point on the line $x$ if $x$ is a line), which can be either left or right of the directed line L.

We now consider the optimality of the above placement function in three singular configurations, shown in Figure 6. In Figure 6(a), $\overleftrightarrow{E}$ is coincident with $\overline{AD}$. In this case, the placement of $E$ anywhere along $\overleftrightarrow{E}$ is optimal and will result in three collinear vertices. The placement function described above will result in the placement of $E$ at either $A$ or $D$ (shown). Figure 6(b) represents the case where three sequential vertices $B$, $C$ and $D$ are collinear. Clearly, optimal placement is at $B$. This is equivalent to removing the middle of the three collinear vertices, in this case $C$. Furthermore, the displacement area associated with this collapse operation will be zero. This means that removal of collinear vertices (including those resulting from the configuration in Figure 6(a)) will be automatically prioritized by the segment collapse algorithm. Lastly, Figure 6(c) shows the situation where $\overline{BC}$ is parallel to $\overline{AD}$ (and thus $\overleftrightarrow{E}$), in other words, $B$ and $C$ are equidistant from $\overline{AD}$. In this configuration,

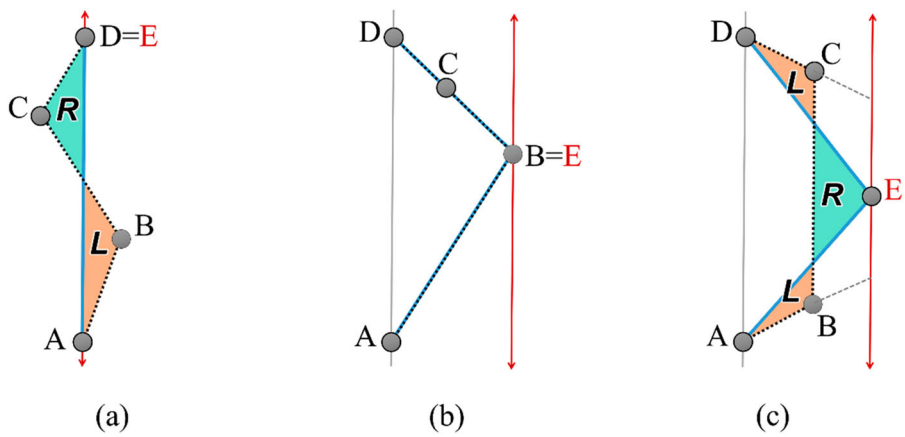


**Figure 6.** Examples of singular configurations, with optimal placement of $E$: (a) $\overleftrightarrow{E}$ is coincident with $\overline{AD}$ (placement anywhere between $A$ and $D$ is optimal, but placement is not necessary), (b) $B$, $C$ and $D$ are collinear (optimal placement is at $B$), (c) $\overline{BC}$ is parallel with $\overline{AD}$ (placement along $\overleftrightarrow{E}$ anywhere between the intersections with $\overline{AB}$ and $\overline{CD}$ is optimal).

placing $E$ anywhere between the locations where $\overleftrightarrow{E}$ intersects $\overline{AB}$ and $\overline{CD}$ will result in equal areal displacement. For each of the above singular configurations, the placement function described above will produce an optimal solution, although in two cases it is just one of many possible such solutions. We note, however, that such singular cases will occur rarely for natural features.

### Preventing self-intersection

We now describe a simple test to avoid self-intersections. First, note that each collapse event in the APSC algorithm results in displacement across at most two triangular regions. Since these triangles may contain more than one new edge, however, the triangle-criterion does not strictly apply. That is, we cannot simply check for the existence of a vertex within these displacement triangles, because a topological error may be caused by a line segment that crosses a triangle but whose endpoints both lie outside of it. However, since the displacement area at each step is well defined it is still possible to check for any topological error caused by this displacement. To prevent self-intersections, it is sufficient to check whether any existing line segment (excluding AB, BC and CD) crosses either AE or ED prior to performing each collapse event and then exclude that collapse operation. By refraining from collapsing any segment in which either of these two triangles intersects an existing segment, it can be guaranteed that no self-intersections will be introduced in the simplification process.

## Comparison of simplification algorithms

### Data

We conducted an experiment to compare the effectiveness of APSC with three well-known line simplification algorithms. As experimental data, a set of ten (10) lakes was selected from different conditions within the conterminous United States. Sample features are from the high-resolution (HR) National Hydrography Dataset (NHD). Within the conterminous United States, HR NHD is a multi-scale dataset composed of hydrographic data from the best available data contributed from source scales of 1:24,000 (24 K) or larger (Stanislawski, Falgout, & Buttenfield, 2015; Stanislawski et al., 2016; U. S. Geological Survey, 2017). Ten sample lakes formed by a variety of geomorphic processes were selected for analysis, with one, two, two, four, and one of the lakes derived from alpine, dam, desert, glacial and karst geomorphic conditions, respectively (Figure 7). The lakes range in area from about 31 to over 63,000 square kilometers, with boundaries containing between 1800 and 410,000 vertices (mean: 54,946, median: 9253). Features are represented with coordinates projected in the North American Albers Equal Area Conic projection.

### Experimental design

Three simplification algorithms – VEA, RDP, and RSM – are compared to the APSC algorithm both with and without a topology check to prevent self-intersections. For comparison purposes, we follow previous researchers (Shi & Cheung, 2006; Tutić & Lapaine, 2009; Visvalingam & Whyatt, 1993; Zhou & Jones, 2005) to simplify each feature to nearly identical numbers of vertices using each algorithm. Since each of the 10 HR NHD lake polylines has a different number of vertices, the target number of vertices to retain for each feature is

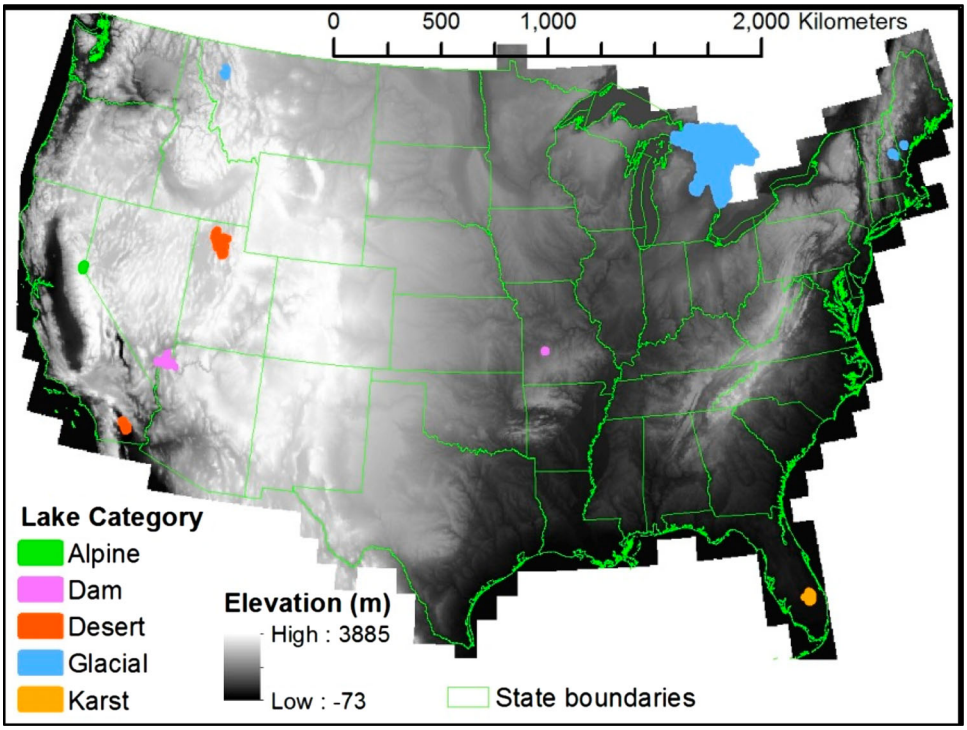


**Figure 7.** Geographic category and distribution of 10 sample lakes in the conterminous United States. Features are selected from the waterbody feature class of the high-resolution National Hydrography Dataset. Five-kilometer resolution data show the approximate range of elevation conditions within the study.

chosen by using the Radical Law (Töpfer & Pillewizer, 1966) to estimate vertices sufficient for target scales of 1:100,000 (100K); 1:500,000 (500 K); 1:1,000,000 (1 M); and 1:5,000,000 (5 M). This results in target proportions of original vertices of 0.49, 0.22, 0.15, and 0.07 respectively. We do not attempt to directly target a given map scale through parameterization because each algorithm uses a different metric for parameterization, so the results would not be comparable. With VEA, RDP and APSC, the target number of vertices is obtained exactly by sequentially removing or adding points. For RSM, we iteratively adjust the controlling parameter (hexagon size) until the target number of vertices is obtained as nearly as possible. In some cases, this results in differences of +/−2 vertices from the target.

### *Assessment*

Several metrics compare the geometric characteristics resulting from each algorithm. Metrics determined for each simplified lake polygon include area, areal displacement, complexity, and modified Hausdorff distance (MHD). The area of original and simplified polygons is computed using the shoelace (Surveyor's) formula (Braden, 1986). For this paper, polygon complexity is defined as the perimeter squared divided by the area, which is the inverse of a commonly used compactness metric (MacEachren, 1985). Areal displacement is computed as the summed area of polygons created when an original

linear feature and a simplified version of the same feature are overlaid. For the case of polygons, the displacement area is determined through a union of the original polygon with its simplified version after all island polygons are removed from the simplified version, which may occur when a boundary intersects itself. The issue of self-intersection is further addressed later in the discussion section.

The Hausdorff distance between two geometric shapes is the largest minimum distance between any point on one shape to any point on the other (Hangouët, 1995; Nutanong, Jacox, & Samet, 2011; Rucklidge, 1996). For computational efficiency, modified Hausdorff distance (MHD) is determined in this paper as the larger of two distances: either the furthest minimum distance between the vertices of the simplified line and its original version, or the furthest minimum distance between the vertices of the original line and its simplified version. Given some vertex reduction, these two distances will not be equal in every case because the minimum distances are not always vertex-to-vertex.

Area, area displacement, complexity, and MHD are computed for original and simplified features for each of the four levels of simplification using each algorithm. Values for 10 lake features are averaged for each level of simplification. Percent change from original values is computed for average complexity and average area. In addition, the maximum MHD is identified among the ten lake features for each simplification level and algorithm.

To assess topological correctness of simplified features, self-intersections are identified on each simplified feature, and the number of intersections is tabulated for each feature. No self-intersections exist in the original lake boundaries.

## Results

Simplification assessment metrics are compared with respect to the percent of original vertices retained for 10 sample features at each level of detail. An average number of vertices decreases with increasing simplification to achieve smaller scale representations, with average vertices of about 55,000 (100%) for original features reducing to about 3800 (7%) at 5 M.

### *Area preservation*

Mean and maximum change in the area of all lakes caused by simplification at the four levels of detail are summarized in Figure 8. When reviewing figures, higher simplification results are on the left moving to less simplification on the right. Results confirm that APSC with topology check results in no net change in area, meaning that displacement areas on the inside and outside of the original features were exactly equivalent. Without the topology check our calculations showed a very slight change in area caused by the APSC, but this may be due to ambiguities of performing calculations on self-intersecting polygons.

Percent absolute change in area (Figure 8(a)) increases on average with increasing levels of simplification for all other algorithms. To illustrate worst-case scenarios, the maximum changes in area for each algorithm were also evaluated. The greatest percent change in area is generated with the RSM algorithm, which causes up to 0.3 percent area change at the 100 K level (49% vertices retained) and 5.6 percent change at the 5 M level (7% vertices retained). Maximum overall change in area at the 5 M level

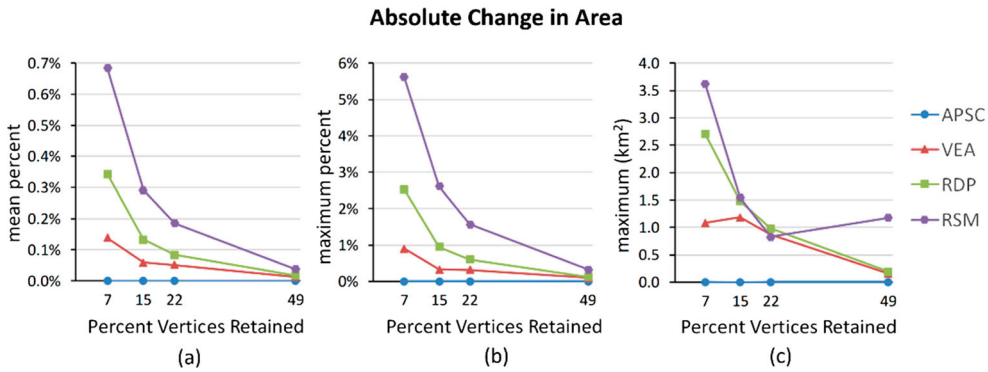


**Figure 8.** Absolute changes in area from original features for 10 lake polygons simplified to 4 levels of detail. (a) mean percent change in area, (b) mean absolute change in area, and (c) maximum absolute change in area. Values for APSC algorithm with and without topology check are nearly identical and cannot be visually distinguished.

ranged from 1.0 km$^2$ for the VEA algorithm to as high as 3.6 km$^2$ with the RSM algorithm. The VEA algorithm generated less area change than RDP at all levels of detail, and RSM generated the most areal change at all but one level of detail.

## *Areal displacement*

Average total displacement area of the simplified features on the inside and the outside of the original features increases with increasing levels of simplification as shown in Figure 9. Average values indicate the APSC algorithm generates the least displacement area at all levels of detail, with a maximum average displacement area of 4.4 km$^2$ at the 5 M level of detail. Including a topology check increased areal displacement slightly (2.5% on average) above the value obtained without the topology check. Increasingly larger average displacement areas are generated by the VEA, RDP, and RSM algorithms, which respectively peak at 5.8, 7.4, and 13.2 km$^2$ at the 5 M level of detail.

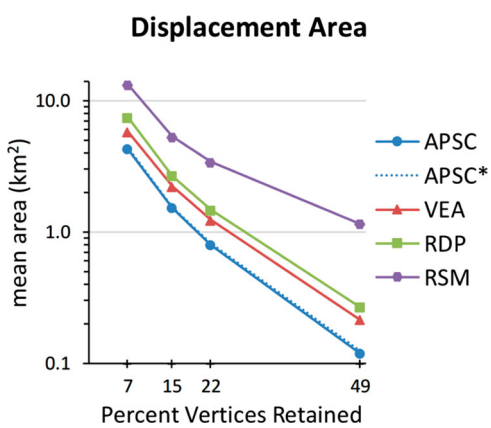


**Figure 9.** Mean displacement area in square kilometers (sq. km.) from original features for ten simplified lake polygons at four levels of detail. Asterisk denotes algorithm implemented with topology check.

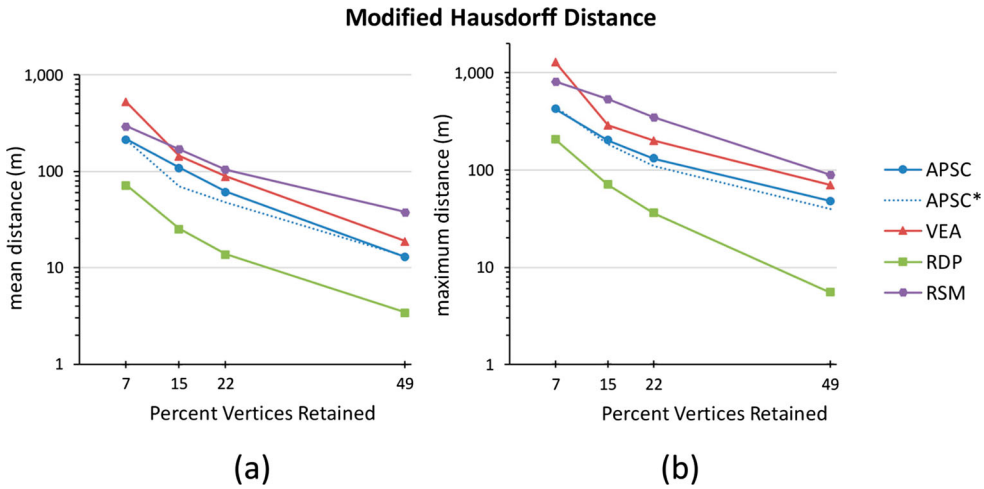


**Figure 10.** Modified Hausdorff distance from original features for 10 simplified lake polygons at four levels of detail. (a) mean modified Hausdorff distance, (b) maximum modified Hausdorff distance. Asterisk denotes algorithm implemented with topology check.

## *Modified Hausdorff Distance (MHD)*

Average and maximum linear displacement measured using MHD values are summarized in Figure 10(a,b), respectively. The RDP algorithm generates the least positional displacement of all algorithms, as indicated by both average and maximum values, with the APSC algorithm generating the next larger values, at all levels of simplification. Shi and Cheung (2006) tested several line simplification algorithms on island boundaries and also found the RDP produced the least positional displacement than other tested algorithms. Inclusion of a topology check reduced MHD noticeably at 500 K and 1 M scales, but this effect varied widely and the topology check also increased MHD for some features. VEA and RSM generate more positional displacement than RDP and APSC at all simplification levels. For the features tested, maximum displacement values from the APSC (both with and without topology check) and RDP are smaller than the National Map Accuracy Standards (NMAS) of 1/50th of an inch (U. S. Bureau of Budget, 1947) at 100 K, 500 K, 1 M, and 5 M, which converts to about 51, 254, 508, and 2540 meters, respectively. VEA creates a maximum MHD value greater than the NMAS standard at 100 K, and RSM creates a value that exceeds standards at 100 K, 500 K and 1 M. Average MHD values, however, were below NMAS limits for all algorithms at all tested scales.

## *Complexity of boundary lines*

Complexity values of the original lakes (as described above) range from 37 to 2475, with an average of 504. The sample lakes are about 3–200 times more complex than a minimum complexity unit circle (complexity = 12.57). Average percent change in feature complexity of the simplified lakes compared to original features is shown in Figure 11. As expected, feature complexity decreases, on average, with increasing simplification in all cases, except for the APSC algorithm, which shows a slight increase in average complexity at

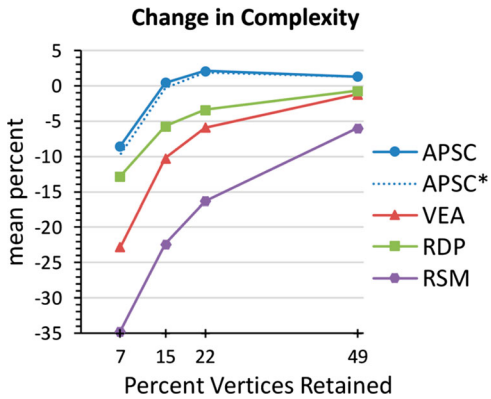


**Figure 11.** Mean percent change in feature complexity from original features for ten simplified lake polygons at four levels of detail. Asterisk denotes algorithm implemented with topology check.

the 100 K, 500 K, and 1 M simplification levels. This pattern was moderated slightly with the inclusion of a topology check. The protection of feature complexity at large scales forms an unexpected result, and indicates that the algorithm tends to extend curves outward more often than it eliminates curves at least at low levels of simplification.

On average RDP and VEA reduce complexity about the same amount as APSC increases it at the 100 K level, with the RSM algorithm having the largest average complexity reduction of about 6 percent. At higher simplification levels, average feature complexity decreases, with APSC, causing the least change, and increasingly larger changes result for RDP, VEA, and RSM, with RSM reducing complexity by nearly 35 percent, on average, at 5 M level of detail. Shi and Cheung (2006) noted the RDP algorithm created the least amount of angular shape distortion among the algorithms tested on island boundaries.

## *Topological correctness*

Prior to incorporating a topology check, the APSC algorithm generated a greater number of self-intersections during simplification processing than did other tested algorithms (Table 1). Self-intersecting lines are, however, generated by all four tested simplification methods and levels of detail, with larger numbers generated in more sinuous parts of

**Table 1.** Self intersections (features with at least one self-intersection) created for 10 simplified lake polygons at four levels of detail using four simplification algorithms.

| | *Simplification level* | | | |
|---|---|---|---|---|
| Algorithm | 0.49 | 0.22 | 0.15 | 0.07 |
| APSC | 4(2) | **67(8)** | **151(9)** | **264(9)** |
| APSC* | 0(0) | 0(0) | 0(0) | 0(0) |
| VEA | 4(2) | 6(3) | 21(6) | 22(6) |
| RDP | 2(1) | 14(6) | 33(8) | 96(6) |
| RSM | **7(3)** | 28(5) | 30(5) | 62(7) |

Notes: Four levels of detail, 1:100,000, 1:500,000, 1:1,000,000, and 1:5,000,000 reduce the number of vertices of each lake feature to about 49, 22, 15, and 7 percent of the vertices of the original feature, respectively. Number of intersections/number of lake boundaries with intersections. Highest values are bold.

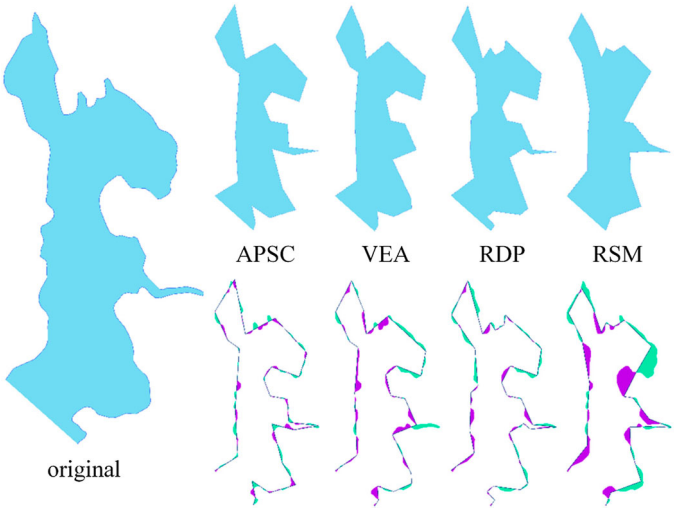


**Figure 12.** A portion of Lake Winnipesaukee, northern New England, with features simplified to 1:1,000,000 scale (top) and associated areal displacement (bottom) for four line simplification algorithms. Areas outside the simplified lines that are eliminated from the lake are shown in green, and areas inside the simplified lines that are added to the lake are shown in purple.

the boundary, and as more vertices are removed. The APSC generates the largest number of intersections at the three smallest levels of detail, creating more than twice as many intersections than the other algorithms respectively. The RSM method generates the most self-intersections at the highest level of detail, 100K. Self-intersections are more common for the APSC algorithm because the introduction of Steiner points can result in exaggeration of existing curves (discussed further below). As noted previously, strategies exist to avoid self-intersection for all line simplification algorithms, but for the present research, a topology check was implemented only for the APSC algorithm. As shown in Table 1, our analysis confirms that the topology check prevented the APSC from introducing any self-intersections.

### *Illustration of the method*

To illustrate the differences between algorithms, a section of Lake Winnipesaukee, a lake of glacial origin in northern New England, is shown in Figure 12 with reductions to the 1M level of detail (15% of vertices retained). All algorithms present a reasonably accurate facsimile of the original feature, but in the displayed region the RDP algorithm retains more points and therefore appears to capture the most detail. However, the APSC algorithm accurately captures important details that are missed by the VEA and RSM algorithms, such as the long and narrow inlet at center right. The RDP algorithm also captures the inlet. Despite the greater detail captured by other algorithms in various parts of the lake, APSC results in the smallest pockets of local areal displacement among the four algorithms (Figure 12, bottom row).

The effects of the topology check are illustrated in Figure 13 for a portion of Lake Mead simplified to the 5 M level of detail (7% of vertices retained). In regions of high-frequency

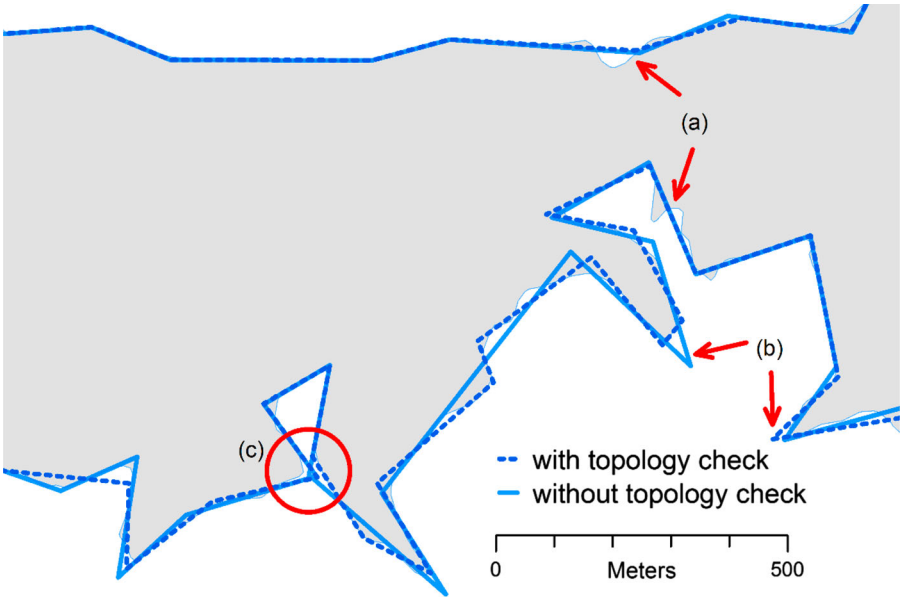


**Figure 13.** A portion of Lake Mead with features simplified to 1:5,000,000 scale using the APSC algorithm with and without performing a topology check to prevent self-intersections. Highlighted regions indicate (a) high-frequency detail, (b) low-frequency detail, (c) a self-intersection that is prevented by the topology check. See text for explanation.

detail where features must be eliminated, APSC tends to find a middle path through inward and outward facing protuberances (Figure 13(a)). On the other hand, in regions of low-frequency detail where large features must be simplified, APSC tends to exaggerate protuberances (Figure 13(b)), which can lead to self-intersection (Figure 13(c)). The topology check avoids the creation of self-intersections, and this avoidance can have a cascading influence on the simplification process elsewhere, especially in the immediate neighborhood of a potential self-intersection.

In a production environment, a typical goal of generalization is to reduce content at smaller scales for data storage and processing requirements while preserving details for legibility or analytic tasks (Gary et al., 2010). In our comparative analyses, the first goal was guided by the Radical Law (Gökgöz, Sen, Memduhoglu, & Hacar, 2015; Stanislawski, 2009; Töpfer & Pillewizer, 1966) to establish an appropriate target number of vertices as a clear and simple standard by which to conduct a controlled comparison of different line simplification algorithms. This approach has been followed by other researchers (Tutić & Lapaine, 2009; Visvalingam & Whyatt, 1992; Zhou & Jones, 2005), but in production environments, it is more common to use a tolerance defined by the largest insignificant detail at target scale. For this purpose, we recommend setting an areal displacement tolerance area equal to the area of a triangle with sides of ¼ mm at the target scale.

Figure 14 illustrates this strategy applied to six lake features in the HR NHD from Minnesota, USA (U. S. Geological Survey, 2017), including lakes characterized by high, medium and low-frequency detail. Simplification to target scales of 1:100 K (4× reduction) and 1:250 K (10× reduction) was performed using the APSC algorithm with areal displacement tolerance equal to 541 and 3383 $m^2$, respectively. This reduced the number of vertices to

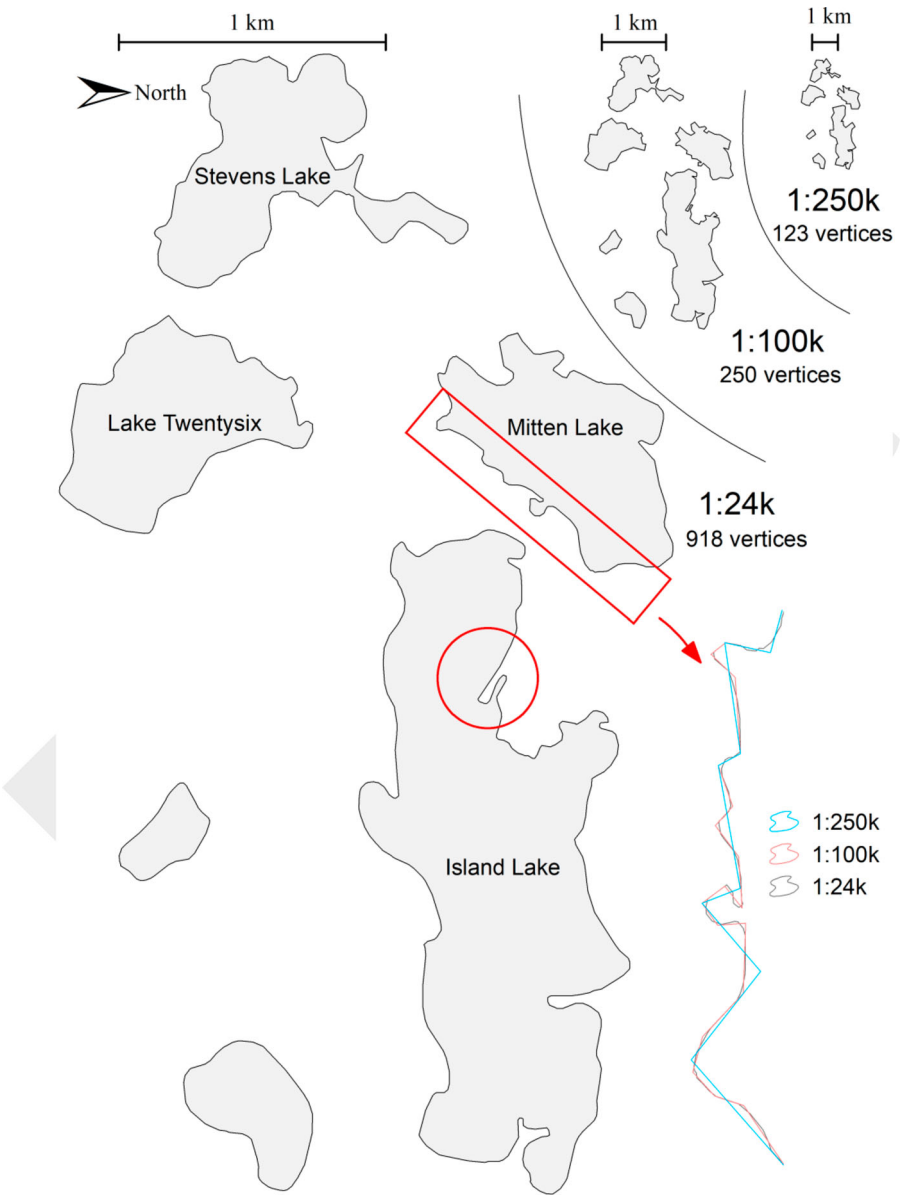


**Figure 14.** APSC algorithm applied to National Hydrology Dataset (NHD) lake boundaries compiled at 1:24 K shown at two target scales (1:100 K and 1:250 K). Each scaled version is labeled with the number of vertices retained. The panel in lower right corner overlays a section of Mitten Lake (Minnesota) boundary at all three scales to demonstrate the amount of spatial coincidence.

27% and 13% of the original count at 1:100 K and 1:250 K, respectively, somewhat fewer than predicted by the Radical Law. When displayed at the target scales, the sub-portions of features remain recognizable and the overlay of a segment of one lake section at all three scales shows close spatial coincidence. Several small angularities are introduced that appear unnaturally sharp in the enlarged overlay, but effectively convey the existence of intrusions and extrusions at target scales. Thus, the method reduces content yet

maintains shape and geometric characteristics such as frequency of detail, preserving global and local polygon shape even across large scale reductions.

## Discussion

This paper presents a simplification algorithm that is novel relative to other simplification methods in wide use today in that it simultaneously aims to preserve area and minimize areal displacement. These two constraints guide the selection and collapse of polyline segments to Steiner points. Whereas most other algorithms that use Steiner points (Cromley, 1992; Li & Openshaw, 1992; Meulemans et al., 2010; Raposo, 2013) may collapse large clusters of vertices locally to a single point at once, the APSC algorithm is incremental, seeking the optimal pair of vertices to collapse at each step. Procedurally, APSC combines some characteristics of VEA (Visvalingam & Whyatt, 1992) and the area-preserving algorithm of Tutić and Lapaine (2009), with the additional development of a procedure for minimizing areal displacement during segment collapse. The algorithm in its current version is computationally feasible for application to large datasets even with the inclusion of topological protection.

Preliminary results indicate the APSC algorithm performs well at tasks for which it was designed and generally outperforms the other algorithms. Specifically, the method precisely preserves area and outperforms the other tested algorithms at minimizing areal displacement. Local pockets of areal displacement are visibly smaller for APSC than for other algorithms, illustrating the advantage of using Steiner points to balance displacement on either side of the polyline. Not surprisingly, the APSC does not perform better than other algorithms at tasks that it was not designed to perform. In tests reported here, the RDP algorithm generates the smallest overall positional displacement at all levels of detail, and substantially outperforms APSC and other tested algorithms in this regard.

The distinctive properties of the APSC algorithm lead to some interesting results. The introduction of Steiner points allows APSC to maintain shape complexity, as bend features appear more likely to be exaggerated than eliminated at early stages of simplification. By the same token, however, in complex or sinuous sections of the features, these exaggerations result in a large number of self-intersections relative to other methods, which distorts topological relations and can impact area computations. This can be remedied with the inclusion of a simple line intersection test to prevent collapse operations that would result in a self-intersection. The self-intersection test had little impact on most quantitative metrics, with the largest difference being a reduction in average and maximum modified Hausdorff distance at some levels of simplification (Figure 10(a,b)). We did not incorporate self-intersection tests for the other algorithms that we compared. However, given that these algorithms resulted in fewer self-intersections to begin with (Table 1), we suspect that such tests would have even smaller impact on the quantitative metrics that were evaluated.

The APSC method appears to maintain horizontal positional displacement within NMAS standards and, on average, maintains smaller horizontal positional changes than the other methods, except for the RDP algorithm as mentioned above. Problems with positional displacement and intersections tend to occur in areas of high-frequency detail, such as along coastlines with embedded docks or piers. This implies the space between the high frequency details where intersections occur are smaller than NMAS standards and possibly

too small to be appropriate for retention at smaller target scales, based upon feature capture specifications or legibility. Some modification of the algorithm may also be warranted to limit the generation of thin or jagged protrusions. While the topological correction eliminates self-crossings, it is a binary operator and as such does not prevent lines from nearly touching each other. Perhaps some additional or alternative generalization operation is needed for proximal repeated or opposing protrusions, for example, the circled segment of the Island Lake polygon in Figure 14. Alternatively, positional displacement constraints could be tightened for overly detailed sections, effectively implementing an adaptive simplification strategy. Several scholars have considered adaptive strategies for line simplification (e.g. Mustiere, 2005; Nakos, Gaffuri, & Mustiere, 2008; Samsonov & Yakimova, 2017; Shahriari & Tao, 2002), and these could be tailored and modified to inform future research on areal displacement.

The comparative testing applied in this analysis constrains features to a specific number of vertices, governed by the Radical Law. This allows for direct comparison among different algorithms but is not the typical parameterization for cartographic simplification algorithms. A more common approach is to designate an acceptable error tolerance (i.e. distance for RDP, hexagon radius for RSM) and permit a more flexible parameterization of the number of vertices to retain. Similar to VEA, the APSC algorithm can be parameterized by designating an acceptable areal displacement at each step. Unlike RDP and RSM however, the error tolerance parameter for both VEA and APSC applies to individual steps only and does not guarantee an upper limit on cumulative error. Consequently, for production scenarios where such a guarantee is required, trial-and-error may be required when using VEA and APSC to determine the appropriate parameter value for each feature.

The APSC algorithm focuses on area preservation and as such is designed specifically for polygon boundary simplification. However, its overall performance makes it a plausible candidate for effectively simplifying any vector line feature. The observed protection of feature complexity, which is distinct from other algorithms, suggests that the algorithm may have unique properties that would make it suitable for specialized applications and this warrants further research. Lastly, the results for APSC reported here all pertain to the specific placement and displacement functions described, but different results might be obtained by substituting different placement and displacement functions into the general segment collapse algorithm.

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